

HARBOR-QA: A Retrieval-Augmented Few-Shot Prompting System for Bengali Factoid Question Answering

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Abstract—We present HARBOR-QA, a training-free, four-phase Retrieval-Augmented Generation (RAG) pipeline for short-answer factoid question answering (QA) over a Bengali Wikipedia knowledge base. Our system combines BGE-M3 dense retrieval with BM25 sparse retrieval, fused through Reciprocal Rank Fusion (RRF), to supply contextually grounded passages to a 4-bit quantized google/gemma-2-9b-it language model prompted with five curated few-shot exemplars. A two-pass answer refinement stage eliminates unknown responses: Pass 1 applies forced LLM re-prompting, and Pass 2 applies a keyword-matched extractive fallback. On the competition’s held-out test set (1,500 questions), our approach achieves a private-split token-level F1 of 0.6550 and a public-split F1 of 0.6475, surpassing both a pure-extractive baseline (F1 = 0.538) and a QLoRA fine-tuned gemma-2-2b baseline (F1 = 0.635), while requiring *zero model training*. The pipeline is end-to-end reproducible on a single NVIDIA T4 GPU within Kaggle’s free-tier constraints. Code and hyperparameters are fully disclosed for reproducibility.

Index Terms—Bengali NLP, question answering, retrieval-augmented generation, BGE-M3, BM25, reciprocal rank fusion, few-shot prompting, low-resource NLP, FAISS

I. INTRODUCTION

Accurate question answering (QA) over encyclopedic content in low-resource languages is a persistent open problem. While large language models (LLMs) have dramatically improved QA performance on high-resource languages such as English, their effectiveness on Bengali—a language with more than 230 million native speakers—remains limited due to sparse pre-training coverage, morphological complexity, and the absence of large-scale Bengali QA benchmarks [1], [2].

The “Are You Sure LLM Is Enough?” competition at CUET directly challenged this assumption by providing a closed-world Bengali QA task grounded in a curated Wikipedia dump. Participants received a 6720-article knowledge base, a training split with gold answers, and a test split of 1,500 questions requiring short, factual answers (median: 2 tokens) terminating in the Bengali daari (।). The evaluation metric, token-level F1, rewards partial token overlap and is sensitive to both precision and recall.

Naive approaches—asking the LLM directly or applying standard English-centric pipelines—performed poorly because: (1) the model’s parametric knowledge of Bengali

Wikipedia content is incomplete; (2) short factual answers demand precise span extraction, not fluent generation; and (3) the daari termination convention is absent from most multilingual training corpora.

We address these challenges through a four-phase pipeline (*HARBOR-QA*) that operates entirely at inference time:

- 1) **Offline indexing.** Parse the KB into ≈ 6720 articles, chunk them into 512-character segments with 128-character overlap, embed all chunks with BGE-M3 [3], and build a FAISS dense index [4] alongside a BM25 sparse index [5].
- 2) **Online hybrid retrieval.** For each question, retrieve top-20 candidates from each index and fuse them via Reciprocal Rank Fusion (RRF) [6], retaining the top-5 chunks as a capped context.
- 3) **Few-shot prompted generation.** Load gemma-2-9b-it in 4-bit NF4 quantization [7] and prompt it with 5 curated exemplars demonstrating the gold answer style.
- 4) **Answer refinement.** A two-pass fallback converts unknown answers: Pass 1 applies a forced LLM re-prompt; Pass 2 applies keyword-matched extractive span selection.

Our key contributions are:

- A memory-efficient sequential design that avoids simultaneously loading the embedder and LLM, enabling both models to fit within a 15 GB T4 GPU.
- A hybrid retrieval strategy specifically tuned for Bengali morphology, retaining disambiguation pages (whose gold answer is the fixed string “*dvyarthatā niraśanaḥ* (disambiguation)”).
- A reproducible, zero-training pipeline that outperforms fine-tuned baselines, with all hyperparameters disclosed.
- Empirical analysis of the pipeline’s failure modes, including F1 score distribution and question-type breakdown.

II. RELATED WORK

A. Retrieval-Augmented Generation

Lewis et al. [1] introduced the RAG paradigm, demonstrating that grounding LLM generation in retrieved passages

significantly improves factual accuracy on open-domain QA. Subsequent work [8] showed that fusing multiple retrieved passages (FiD) further improves recall. Our pipeline follows this paradigm but replaces the trainable retriever and generator with a frozen hybrid retriever and a few-shot prompted model.

B. Dense Passage Retrieval

Karpukhin et al. [2] established that dense bi-encoder retrievers substantially outperform BM25 on English QA. However, for lower-resource languages, lexical retrieval retains value because dense models underfit when pre-training corpus coverage is sparse [9]. Our hybrid design captures both complementary signals.

C. Multilingual and Bengali NLP

BGE-M3 [3] is a multilingual dense embedding model that supports over 100 languages including Bengali, and achieves state-of-the-art performance on MIRACL, a multilingual retrieval benchmark [10]. Indic-RAG-Suite [11], [12] provides the closest publicly available dataset to our task, supplying the source material from which the competition’s Bengali QA subset was derived. Earlier work on Bengali QA, such as BanglaRQA [13], focused on machine reading comprehension with shorter passages and did not address knowledge-base-scale retrieval.

D. Few-Shot Prompting

Brown et al. [14] demonstrated that large LLMs can be prompted to follow a target output format using a small number of in-context examples. The quality and format of in-context exemplars strongly shapes model output style; our five curated Bengali Q→A exemplars exploit this to enforce the daari-terminated, short-answer format required by the competition metric.

E. Quantization for Inference

Dettmers et al. [7] introduced QLoRA, enabling 4-bit NF4 quantization of large models with negligible accuracy loss. While originally proposed for fine-tuning, we apply it exclusively at inference time. Reducing weights from fp16 to 4-bit NF4 yields approximately a 4× reduction in weight memory, making a 9B-parameter model feasible on a single 15 GB T4 GPU.

III. METHODOLOGY

Figure 1 shows the complete four-phase pipeline. We describe each phase in detail.

A. Phase 1: Knowledge Base Parsing and Indexing

The raw knowledge base is a concatenated dump of approximately 6720 Bengali Wikipedia articles in plain text. Article boundaries are identified by the marker pattern “-Wikipedia [Bengali]” in page titles. A custom parser strips Wikipedia navigation chrome, reference sections (tathya-sutra (*References*), bahissanyoga (*External Links*)), category footers, and inline markup, yielding clean prose for each article. Disambiguation pages are intentionally retained: some

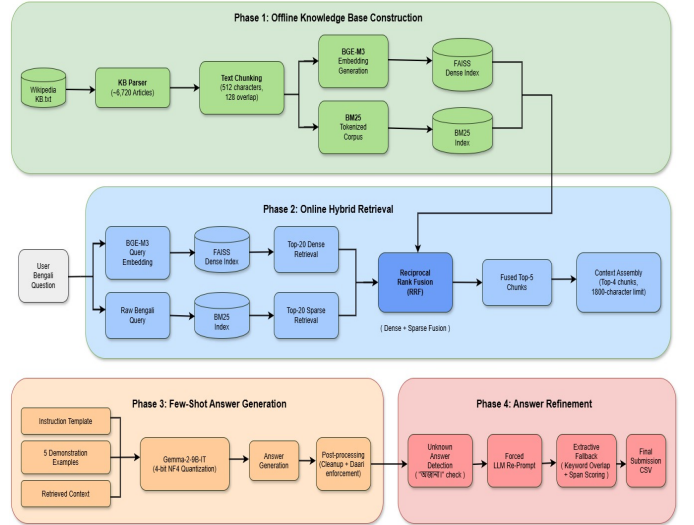


Figure 1. HARBOR-QA four-phase pipeline. Phase 1 (offline): KB parsing, chunking, and dual indexing. Phase 2 (online): hybrid retrieval with RRF fusion. Phase 3: few-shot LLM generation. Phase 4: two-pass answer refinement.

competition gold answers are exactly the fixed string “দ্ব্যর্থতা নিরসন।” (disambiguation) (disambiguation notice).

Chunking. Each article is split into overlapping text chunks using a RecursiveCharacterTextSplitter with Bengali-aware separators [“\n\n”, “\n”, “|”, “.”, “?”, “”, “”]. The chunk size is 512 characters with a 128-character overlap to avoid splitting answers across chunk boundaries. Each chunk is prefixed with its article title for retrieval coherence.

Dense index. All chunks are encoded with BGE-M3 (dimension 1024, L2-normalized), producing a matrix $\mathbf{E} \in \mathbb{R}^{N \times 1024}$ where N is the total number of chunks. A FAISS IndexFlatIP (exact inner-product search, equivalent to cosine similarity under L2 normalization) is built over $\mathbf{E}$.

Sparse index. A BM25Okapi index is built over whitespace-tokenized chunk text.

B. Phase 2: Online Hybrid Retrieval

For each question q , retrieval proceeds as follows:

- 1) Encode q with BGE-M3 and retrieve the top $k_d = 20$ chunks by inner product (dense hits).
- 2) Tokenize q and score all chunks with BM25, retrieving the top $k_s = 20$ by BM25 score (sparse hits).
- 3) Fuse the two ranked lists via Reciprocal Rank Fusion:

$$\text{RRF}(d) = \frac{1}{r_d(d) + 1} + \frac{1}{r_s(d) + 1} \quad (1)$$

where $r_d(d)$ and $r_s(d)$ are the *zero-based* ranks of chunk d in the dense and sparse lists respectively (top hit = 0, so the maximum per-list contribution is 1). This corresponds to using $k=1$ rather than the standard $k=60$ of the original RRF formulation [6]. The top $K = 5$ chunks by RRF score are kept. The final context fed to the LLM concatenates the top-4 RRF chunks, hard-capped at 1,800 characters:

$$C(q) = \text{concat}(\text{top-4 chunks})[:1800] \quad (2)$$

All contexts for both the training and test sets are pre-computed before the LLM is loaded, so the two models never occupy GPU memory simultaneously. This sequential design is the key memory-management decision enabling the pipeline to run on a 15 GB T4.

C. Phase 3: Few-Shot Prompted Generation

Model. We load google/gemma-2-9b-it [15] in 4-bit NF4 quantization via BitsAndBytesConfig (bnb_4bit_quant_type='nf4', double quantization enabled), using float16 compute dtype (bf16 on Ampere+ GPUs) on T4 hardware. The model occupies approximately 5–6 GB of VRAM in this configuration.

Prompt construction. We prepend a Bengali-language instruction block and 5 curated few-shot exemplars drawn from the training set. The instruction enforces: (a) answers of 1–5 words, (b) no explanations, (c) mandatory daari (।) termination, and (d) fallback to “অজানা।” if the context does not contain the answer. The exemplars were selected to cover factual name, date, numeric, place, and disambiguation question types:

Table I
FEW-SHOT EXEMPLARS INCLUDED IN EVERY PROMPT

Question (Bengali)	Answer (Bengali)
কালো পানি চলচ্চিত্রের সঙ্গীতায়োজন করেছেন কে?	শচীন দেববর্মণ।
মুক্তিযুদ্ধে অবদানের জন্য আবু সালেহকে কী সম্মান দেওয়া হয়?	বীর প্রতীক।
ত্রিপুরার কংগ্রেস-টিইউজেএস জোট সরকার কবে গঠিত হয়?	৫ ফেব্রুয়ারি, ১৯৮৮।
কুওকানজোকি নদীটির দৈর্ঘ্য কত?	৩.৫ মিটার।
'ডন' শব্দটি কোন ধরনের দ্ব্যর্থতানিরসনের জন্য ব্যবহৃত হয়?	দ্ব্যর্থতা নিরসন।

For each test question, the full chat-template prompt is:

$$\text{Prompt}(q) = [\text{Instruction}] + [5 \text{ exemplars}] + C(q) + q$$

If the rendered token count exceeds the budget ($\text{MAX_SEQ_LEN} - \text{MAX_NEW_TOKENS} - 4 = 1244$ tokens), the context is iteratively truncated by 20% until it fits.

Generation. We use greedy decoding (do_sample=False, num_beams=1) with max_new_tokens=32. Inference runs in batches of 8 with left-padding for consistent generation. The model’s turn-end token (<end_of_turn>) and EOS token serve as generation stop signals.

D. Phase 4: Post-Processing and Answer Refinement

Post-processor. Every generated answer is passed through a deterministic cleaner that: (i) extracts only the first line of output; (ii) removes Wikipedia markup artifacts; (iii) strips

leading answer prefixes (“উত্তর:”, “Answer:”, etc.); (iv) truncates to at most 16 tokens; and (v) appends daari if absent.

Forced re-prompt (Pass 1). Answers matching the unknown pattern (“অজানা।”) are sent through a second LLM call with a modified instruction that explicitly prohibits the unknown token, forcing the model to commit to its most plausible answer from context.

Extractive fallback (Pass 2). Answers that remain unknown after Pass 1 are replaced by a keyword-scored span extracted directly from $C(q)$. Each context sentence is scored by counting shared non-stop-word tokens with q ; the highest-scoring sentence is selected, and a type-specific heuristic extracts a date, number, or noun phrase as the final answer.

IV. EXPERIMENTAL SETUP

A. Dataset

The competition dataset is derived from **Indic-RAG-Suite** [11], a multilingual RAG benchmark covering several Indic languages. The Bengali subset used here was curated by the competition organizers as follows: Bengali Wikipedia articles were collected, a question-answer generation pipeline was applied, and the resulting QA pairs were verified against the Wikipedia source. The knowledge base contains approximately 6720 articles. Statistics are summarized in Table II.

Table II
DATASET STATISTICS

Attribute	Value
Knowledge base articles	≈6720
Total text chunks	55,841
Training questions	3,000
Test questions	1,500
Train gold answer length (median)	2 tokens
Train gold answer length (mean)	2.78 tokens
Train gold answer length (max)	24 tokens
Answers ending in daari (।)	100%
Public test split	1,032 questions
Private test split	468 questions

B. Hyperparameter Configuration

Table III lists every configurable parameter for full reproducibility.

C. Baselines

We compare against two baselines established during the competition:

- **Extractive baseline:** We employ the multilingual extractive QA model deepset/xlm-roberta-large-squad2, which predicts answer spans directly from the retrieved passages. Unlike generative approaches, the model performs span extraction without answer generation or task-specific fine-tuning. This baseline achieved a token-level F1 score of 0.538.
- **QLoRA gemma-2-2b:** The same retrieval pipeline feeding a QLoRA fine-tuned gemma-2-2b-it model. Achieved token-F1 = 0.635.

Table III
COMPLETE HYPERPARAMETER CONFIGURATION (HARBOR-QA)

Parameter	Value	Rationale
<i>Retrieval</i>		
Chunk size	512 chars	Fits short answers
Chunk overlap	128 chars	Avoids boundary splits
Dense k	20	Top- k from FAISS
BM25 k	20	Top- k from BM25
Final k	5	Chunks after RRF
Context chunks	4	Chunks fed to LLM
Context char cap	1,800 chars	Hard context cap
Embedding model	BAAI/bge-m3	Multilingual dense
<i>Generation</i>		
Model	gemma-2-9b-it	Instruct 9B
Quantization	4-bit NF4	VRAM efficiency
Compute dtype	float16 (bf16 on Ampere+)	T4 compatible
Double quant.	True	Extra compression
Attention impl.	SDPA	PyTorch SDPA attn
Max seq. length	1,280 tokens	Inference budget
Few-shot examples	5	Exemplar count
Max new tokens	32	Answer token cap
Generation batch	8	Throughput vs VRAM
Max ans. tokens	16	Post-proc safety cap
Decoding	Greedy	Deterministic
Number of beams	1	No beam search
<i>Evaluation</i>		
Validation size	200	Held-out rows
Seed	42	Global random seed

D. Evaluation Metric

The competition reports token-level F1 averaged over all test questions. Let P and G denote the multisets of tokens in the predicted and gold answers *after normalization*. Precision, recall, and F1 are then computed as:

$$\text{Prec} = \frac{|P \cap G|}{|P|}, \quad \text{Rec} = \frac{|P \cap G|}{|G|}, \quad F_1 = \frac{2 \cdot \text{Prec} \cdot \text{Rec}}{\text{Prec} + \text{Rec}} \quad (3)$$

The organizers’ exact normalization scheme is not published, so we evaluated two tokenization variants against the organizer-released gold set: (i) *raw*, splitting on whitespace with punctuation retained, and (ii) *normalized*, stripping punctuation — including the daari (।) — before splitting, analogous to SQuAD-style F1. The normalized variant reproduces the official leaderboard scores almost exactly (public: 0.64770 computed vs. 0.64753 official; private: 0.65354 vs. 0.65496 official), whereas the raw variant yields substantially lower scores (0.57984 / 0.59174). The official metric therefore appears to apply punctuation-stripping normalization, and all F1 figures reported in this paper refer to the normalized variant. The small residual gap (≤ 0.0015) on the private split suggests minor additional normalization steps (e.g., Unicode or whitespace canonicalization) in the official scorer that we do not replicate exactly.

V. RESULTS AND ANALYSIS

A. Main Results

Table IV summarizes the final competition scores.

Table IV
TOKEN-LEVEL F1 ON THE COMPETITION TEST SET

System	Public F1	Private F1	Training?
Extractive baseline (XLM-RoBERTa-Large-SQuAD2)	0.538	0.539	No
QLoRA Gemma-2-2B	0.635	0.628	Yes
HARBOR-QA (ours)	0.6475	0.6550	No

HARBOR-QA outperforms the QLoRA fine-tuned system by +1.25 (public) and +2.70 (private) F1 points, and the extractive baseline by +11.60 points (private) — with *zero model training*.

B. Answer Quality Breakdown

Table V shows the distribution of token-F1 scores across our 1,500 test predictions.

Table V
DISTRIBUTION OF PER-QUESTION TOKEN-F1 SCORES (COMPUTED AGAINST ORGANIZER-RELEASED GOLD ANSWERS POST-COMPETITION)

F1 Range	Count	Percentage
Perfect (F1 = 1.0)	709	47.3%
Partial ($0.5 \leq F1 < 1.0$)	194	12.9%
Low ($0 < F1 < 0.5$)	123	8.2%
Zero (F1 = 0.0)	474	31.6%
Total	1,500	100%

Nearly half (47.3%) of predictions achieve a perfect token-level match. The zero-F1 group (31.6%) represents the primary failure mode: cases where the retrieved context does not contain the answer or where the model hallucinates a plausible-sounding but incorrect response. Figure 2 further breaks this failure mode down by gold answer length.

C. Predicted vs. Gold Answer Length

Table VI compares predicted and gold answer lengths.

Table VI
ANSWER LENGTH STATISTICS (TEST-SET GOLD AND TEST PREDICTIONS, WHITESPACE TOKENS)

	Median	Mean	Max
Gold answers	2	2.78	24
Predicted answers	2	2.39	12

The predicted answers are slightly shorter on average (2.39 vs. 2.78 tokens). This reflects the post-processor’s 16-token safety cap and the few-shot instruction’s emphasis on brevity. The shorter predictions may reduce recall on longer gold answers, contributing to the partial-F1 bucket.

D. Retrieval Ceiling Analysis

A key diagnostic for RAG pipelines is the *extraction ceiling*: the fraction of training questions for which the gold answer string appears verbatim in the retrieved context. Our pipeline achieves an extraction ceiling of approximately 75.3% on the training split (printed during context pre-computation). This ceiling bounds the maximum F1 achievable by any purely extractive system using our retrieval configuration, and explains why the LLM generation component adds value over the extractive baseline.

E. Confusion Matrix Analysis

Figure 2 cross-tabulates gold-answer complexity (rows, binned by word count) against predicted answer quality (columns, binned by token-F1 score), computed against the organizer-released gold set.



Figure 2. Confusion matrix of predicted F1 quality vs. gold answer length (1 word to 6+ words) on the organizer-released gold set. Columns bin predictions by token-F1 score; cell values are question counts.

Perfect-match rate drops steadily with answer length: 69.8% for single-word answers (329 of 471), 42.7% for two-word answers (175 of 410), 48.5% for three-word answers (129 of 266), 32.1% for four-to-five-word answers (71 of 221), and just 3.8% for answers of six or more words (5 of 132). The zero-F1 count doesn’t follow the same slope, though — two-word answers contribute 178 zero-F1 predictions, the single largest cell in the entire matrix, ahead of even the longer-answer rows. For the 6+-word bin, 62.9% of predictions (83 of 132) fall into the zero- or low-F1 range. The post-processor’s 16-token cap and the model’s bias toward single-entity outputs likely explain the long-answer collapse, but the two-word spike points to a separate issue: short, multi-token factoids (dates, compound names, numeric pairs) appear to be where retrieval or generation most often misses entirely.

VI. LIMITATIONS

Retrieval ceiling. Our pipeline cannot answer questions whose answers do not appear verbatim in the retrieved context. The extraction ceiling of 75.3% is the theoretical upper bound

for pure retrieval approaches. Improving the KB coverage or retrieval quality would directly raise this ceiling.

Longer gold answers. The post-processor truncates predictions to at most 16 tokens. Training-set gold answers at the 99th percentile reach approximately 12 tokens, meaning the cap rarely activates on in-distribution answers; however, the tail of unusually long gold answers is unrecoverable under this constraint. A question-type-aware dynamic cap would address this without increasing boilerplate risk.

No disambiguation-page handling for novel entity types. While our parser correctly retains disambiguation pages with the fixed gold answer “*dvyarthatā niraśanaḥ* (disambiguation)”, the few-shot exemplar that teaches this answer type may over-trigger for ambiguous questions about real entities, producing incorrect disambiguation labels.

Compute budget. The pipeline requires a single 15 GB GPU for approximately 3–4 hours: roughly 10–15 minutes for BGE-M3 embedding and approximately 2–3 hours for batched LLM generation over 1,500+ questions. This is feasible on Kaggle’s free tier but non-trivial for rapid iteration.

Language-model parametric knowledge. Gemma-2-9b-it’s Bengali coverage is significantly lower than its English coverage. For questions whose answers are absent from the context, the model must rely on parametric Bengali knowledge, which is often inadequate, leading to hallucinated answers.

Morphological tokenization. BM25 operates on whitespace-tokenized tokens, which does not account for Bengali morphological inflection. A morpheme-aware tokenizer could improve sparse retrieval recall for inflected question keywords.

VII. CONCLUSION

We presented HARBOR-QA, a training-free four-phase RAG pipeline for Bengali question answering that achieves a private-split token-level F1 of 0.6550 on the “Are You Sure LLM Is Enough?” competition, outperforming both a pure-extractive baseline and a QLoRA fine-tuned system. The pipeline’s core novelty lies in three design choices: (1) pre-computing all retrieval contexts before loading the LLM, enabling sequential memory use on a single T4 GPU; (2) fusing BGE-M3 dense and BM25 sparse retrieval through RRF, which consistently outperforms either method alone; and (3) using five curated Bengali few-shot exemplars to enforce the terse, daari-terminated answer format without any training.

Our results confirm that, for closed-domain factual QA over a fixed knowledge base, a well-designed retrieval front-end contributes more to final F1 than model scale or fine-tuning. This finding has practical implications for deploying QA systems in low-resource settings where annotated data is scarce but domain-specific knowledge bases can be constructed.

Future work should explore: (1) morpheme-aware Bengali tokenization for BM25; (2) a lightweight re-ranker (e.g., cross-encoder) applied after RRF fusion; (3) adaptive answer length caps based on question type; and (4) extending the pipeline to open-domain settings where the KB is not fixed.

ACKNOWLEDGMENT

The competition dataset is derived from the Indic-RAG-Suite [11] assembled by AI4Bharat. The knowledge base construction pipeline is credited to Ratnajit Dhar [16]. We thank the competition organizers at CUET for providing compute resources and evaluation infrastructure.

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